

Calculation of Unbalanced Power Flow in Distribution Networks with Photovoltaic Systems

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Abstract—The existing symmetric component analysis methods cannot provide accurate results and cannot model networks under unbalanced conditions. This article proposes an improved algorithm for calculating the power flow of unbalanced distribution networks containing photovoltaic power based on the forward and backward sweep method. This algorithm has good adaptability to the power flow calculation of distribution networks containing photovoltaic systems. This paper comprehensively analyzes the power flow calculation in three-phase unbalanced distribution networks containing photovoltaic power generation, and points out that the forward and backward sweep method has significant advantages in calculating power flow in distribution networks based on the essential characteristics of each method. A power flow calculation method for unbalanced distribution networks containing new energy sources such as photovoltaic is proposed based on the forward and backward sweep algorithm. The effectiveness of the proposed method is demonstrated through numerical simulations on a test distribution network that includes distributed generation. Results show the improved accuracy and efficiency of the proposed method.

Keywords—unbalanced distribution networks, power flow calculation, photovoltaics, back-forward sweep algorithm

I. INTRODUCTION (HEADING 1)

At present, distributed photovoltaic technology has been increasingly applied, and a large number of photovoltaics have been connected to the power grid. Due to the randomness of photovoltaic output, it brings difficulties to the operation and scheduling of distribution networks containing photovoltaic energy. As a fundamental part of power system analysis, power flow calculation is widely used in many distribution system control scenarios. Therefore, an efficient and practical calculation method is needed to better meet the operation and scheduling of photovoltaic distribution networks.

The actual distribution network situation is complex, usually three-phase unbalanced. In order to accurately calculate the voltage and current in the distribution network, this article analyzes the advantages and disadvantages of several power flow calculation methods for asymmetric distribution networks with photovoltaic access, and uses the improved forward and backward sweep method to calculate the tidal current distribution of the system.

II. POWER FLOW METHODS ANALYSIS

Traditional distribution networks include source nodes and other load nodes, where the source node knows the voltage amplitude and phase angle, while other nodes know the active and reactive loads. The solution is based on the

voltage amplitude and phase angle of other nodes. The known quantities for the corresponding distribution network power flow calculation are network topology, line impedance, and active and reactive loads of nodes, and the solving quantity is node voltage.

The first two known quantities can be represented by the node admittance matrix. For an N-node power system, if the network structure and component parameters are known, the network equation can be expressed as

$$Y\dot{V} = \dot{I} \quad (1)$$

In the formula, Y is $N \times N$ Node admittance matrix, $\dot{V}$ is the node voltage column vector, $\dot{I}$ is the column vector of node load current.

The relationship between node load current and node load power is as follows

$$\text{diag}\{\hat{V}\}\dot{I} = \hat{S} \quad (2)$$

$\text{diag}\{\hat{V}\}$ is a diagonal matrix composed of conjugate node voltages, $\hat{S}$ is the conjugate of node load power.

From the above two equations, it can be concluded that

$$\hat{S} = \text{diag}\{\hat{V}\}Y\dot{V} \quad (3)$$

This is the complex form of the power flow equation, which is an N-dimensional nonlinear complex algebraic system.

$$P_i - jQ_i = \hat{V}_i \sum_{j=1}^N Y_{ij} \dot{V}_j \quad i = 1, 2, \dots, N \quad (4)$$

By introducing the node voltage in the form of complex Cartesian or polar coordinates, further derivation of the power flow equation can be carried out. The basic form of the power flow equation is shown in equation (3). This equation is a nonlinear equation that cannot be directly solved mathematically and cannot be accurately solved through a single matrix operation. Different algorithms can only be used to solve the voltage amplitude from an initial value. At present, the known unknowns of various power flow calculation methods are the same, and the calculations are basically based on equations (1) and (2).

The difficulty of power flow calculation and the work of various methods is to approximate the above equations (1) and

(2) into linear solvable equations, and the voltage and current are coupled with each other through equations (1) and (2).

Gauss method and Newton method are both mathematical methods for solving nonlinear equations. Gauss method utilizes the idea of fixed point iteration, by iterating to calculate the current in equation (2) first, a linear iterative equation for calculating voltage is obtained. Newton method calculates the Jacobian matrix of the point based on the solution of each iteration, essentially achieving linearization by calculating the first derivative. The forward and backward sweep method also separates the process of calculating voltage in equation (1) and current in equation (2) through iteration, similar to the Gaussian method. However, large-scale matrix solutions are not used in the calculation, and they are solved in the order of forward and backward sweep in the distribution network. According to the reference[1], in the case of an acyclic network, this algorithm is another expression of the Z-matrix Gaussian method.

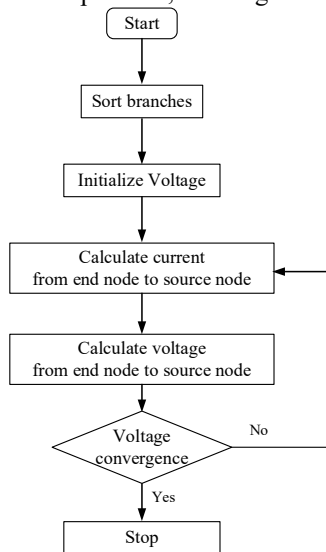
The Gauss method uses the node admittance matrix Y_{bus} instead of explicitly forming the node impedance matrix Z_{bus} , thus becoming the implicit Z_{bus} Gaussian method. The principle of this method is to assume that the voltage of the source node (power node, root node) is a constant value, and the load node injects a constant current. The voltage of each node is calculated through matrix operation. Describe the distribution network using node admittance matrix Y_{bus} , current vector I , and voltage vector V :

$$I = Y_{bus} V$$

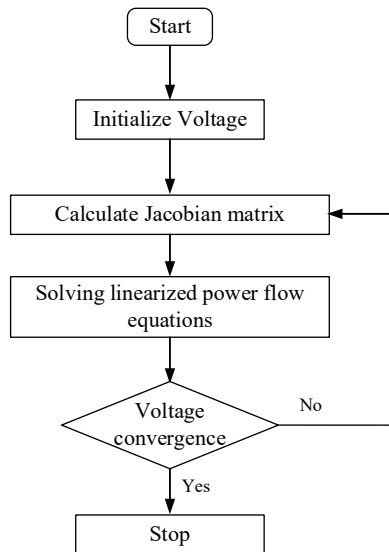
$$\begin{bmatrix} I_s \\ I_j \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \begin{bmatrix} V_s \\ V_j \end{bmatrix} \quad (5)$$

The convergence performance of the Gaussian method depends on the voltage nodes in the network. If there are more voltage nodes in the system, the convergence speed slows down. The calculation example in the reference[2] points out that after the IEEE33 node system is merged into a PV node, the Gaussian method cannot converge. The idea of the article is to approximate the PV node and convert it into a PQ node by injecting reactive power.

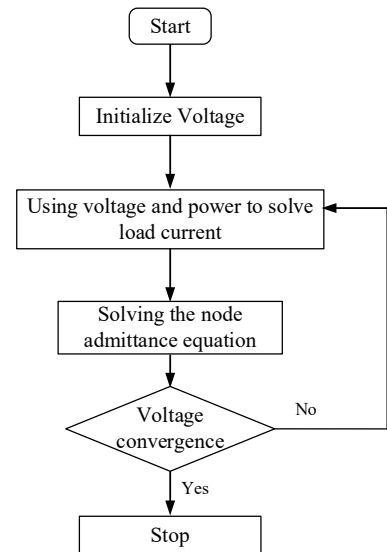
The direct method represents the load of each node with a constant impedance, forming a loop from the feeder source



(a) forward and backward sweep method



(b) Newton method



(c) Gauss method

Fig. 1. Calculation process of each method

node to each load node. The loop current is used as a variable and the loop current equation is listed based on KVL. Using LU decomposition to operate on the equation, after a certain number of iterations, the load current and node voltage can be obtained. The loop impedance method has excellent convergence reliability and stability, and has strong loop processing ability, and the increase of loops does not affect the convergence performance. After improving the algorithm, this method can also calculate three-phase power flow.

The forward and backward sweep method first calculates the load current of the node through equation (2) in one iteration, and then solves for the voltage drop on each branch in the order of the branches. This method has a first-order convergence rate and is relatively simple to calculate.

Reference[3] conducted a theoretical study on the convergence rate of the Newton method, demonstrating that the forward and backward extrapolation method has a fast convergence similar to the strict Newton method when used in the power flow calculation of radial distribution networks with constant power loads. For constant power loads, the difference between the forward and backward sweep method and the Newton method is that the Newton method considers the nonlinearity of branch network losses, while the approximate partial derivative of branch network losses on branch power in the Newton method is very small compared to the non-zero value in the Jacobian matrix. When considering the voltage static characteristics of the load, the forward and backward sweep method makes more approximations than the Newton method, and the convergence speed will decrease.

From an algorithmic perspective, the Newtonian Jacobian matrix with branch complex power as the state variable has the characteristic of upper trigonometry in each sub matrix after slight approximation, which allows the strict solution of the Newton method iterative equation to be directly solved by a recursive formula, which is exactly the full differential form of the forward calculation formula in the forward and backward calculation method. The fundamental reason for the good convergence of the forward and backward sweep method is the radial characteristics of the distribution network.

TABLE I. ADVANTAGES AND DISADVANTAGES OF DIFFERENT METHODS

Methods	Advantageous	Disadvantages
Gauss method	The principle is simple, occupying less memory and computation.	Slow convergence speed and easy non convergence.
Newton method	Fast convergence speed, high convergence reliability, and the ability to handle ring networks.	High requirements for initial values
Forward and backward sweep method	Suitable for radial distribution networks with fast calculation speed.	Ability of handling ring networks is lack.
Direct method	It can calculate the ring network and occupies less memory.	Too slow calculation speed.

III. IMPROVED FORWARD AND BACKWARD SWEEP METHOD

A. Calculate unbalanced power flow

For the simple distribution network shown in the figure below, node i is the root node, and the impedance matrices of i - j and j - k branches are all modeled using an asymmetric model, while the load of j - k nodes is modeled using an asymmetric model.

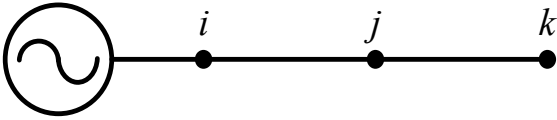


Fig. 2. Simple distribution network

The steps for calculating unbalanced three-phase power flow are similar to that of balanced power flow. Firstly, the three-phase current of the line is calculated from the end node. The current of the end node is the load current, and the current of other nodes is the sum of the current flowing to the next node and the load current. The process of the n th forward iteration is:

$$\begin{bmatrix} i_{ka} \\ i_{kb} \\ i_{kc} \end{bmatrix}^n = \begin{bmatrix} \left(\frac{\dot{S}_{ka}}{\dot{U}_{ka}^{n-1}} \right)^* \\ \left(\frac{\dot{S}_{kb}}{\dot{U}_{kb}^{n-1}} \right)^* \\ \left(\frac{\dot{S}_{kc}}{\dot{U}_{kc}^{n-1}} \right)^* \end{bmatrix} \quad (6)$$

$$\begin{bmatrix} i_{ja} \\ i_{jb} \\ i_{jc} \end{bmatrix}^n = \begin{bmatrix} i_{ka} \\ i_{kb} \\ i_{kc} \end{bmatrix}^n + \begin{bmatrix} \left(\frac{\dot{S}_{ja}}{\dot{U}_{ja}^{n-1}} \right)^* \\ \left(\frac{\dot{S}_{jb}}{\dot{U}_{jb}^{n-1}} \right)^* \\ \left(\frac{\dot{S}_{jc}}{\dot{U}_{jc}^{n-1}} \right)^* \end{bmatrix} \quad (6)$$

The backward iteration process is:

$$\begin{bmatrix} \dot{U}_{ja} \\ \dot{U}_{jb} \\ \dot{U}_{jc} \end{bmatrix}^n = \begin{bmatrix} \dot{U}_{ia} \\ \dot{U}_{ib} \\ \dot{U}_{ic} \end{bmatrix}^n - \begin{bmatrix} Z_{aa} & Z_{ab} & Z_{ac} \\ Z_{ba} & Z_{bb} & Z_{bc} \\ Z_{ca} & Z_{cb} & Z_{cc} \end{bmatrix} \begin{bmatrix} i_{ja} \\ i_{jb} \\ i_{jc} \end{bmatrix}^n \quad (7)$$

The remaining calculation steps are consistent with the balanced power flow calculation.

B. Calculate power flow with photovoltaic generations

At present, research on the flow calculation of new energy integration usually considers the following models: PI node, PQ node, PV node, PQ (V) node. Among them, distributed power sources such as photovoltaic, some wind turbines, fuel cells, and energy storage systems connected to the grid through current controlled inverters usually use the PI node model. The calculation methods of various node models are analyzed as follows:

PI type nodes can be regarded as constant current loads, and this method has strong adaptability to the forward and backward sweep method used in this article.

PQ type nodes can be regarded as equivalent loads, which are consistent with the properties of conventional nodes in power flow calculations.

PV and PQ (V) type nodes still need to consider node voltage when calculating power. Existing algorithms, except for the root node voltage, all other node values and voltages are unknown and cannot be directly given. Therefore, methods such as reactive power injection can be used.

The calculation steps for the forward and backward sweep method are below:

- 1) 1) Sort the branches and initialize the voltage.
- 2) 2) Calculate the current from the end node: If it is a load node, use PQ and V to calculate the load current I . If it is a PI node, add the current directly.
- 3) 3) Calculate the voltage from the root node: Calculate the voltage of the sub nodes through line voltage and voltage drop.
- 4) 4) Stop: If the calculation does not converge, start from step 2 until the calculation converges.

For PI type nodes in photovoltaic systems, the forward and backward sweep method has good adaptability. The forward and backward sweep method can establish the coupling relationship between V-PQ (or V-I) for iterative solution, and the current of PI nodes can be involved in the calculation of branch current iteration through simple addition. In contrast, the Newton method does not include the PI node type and only considers the variable V , δ , P and Q . Establishing relevant equations is needed.

$$P^2 + Q^2 = |I|^2(e^2 + f^2), V = e + jf \quad (8)$$

The Jacobian matrix will end up increasing as the number of equations increases, therefore the difficulty of using Newton's method for calculation will increase.

IV. TEST CASES AND RESULTS

A 9-node single power supply testing system is adopted, with a standard voltage of 1 p.u. and asymmetric parameters for both the line and load. In order to highlight the asymmetry of the results, the parameters of each branch line and the load of each node adopt the same three-phase asymmetric parameters, and the active load of phase B is greater than that of phase AC.

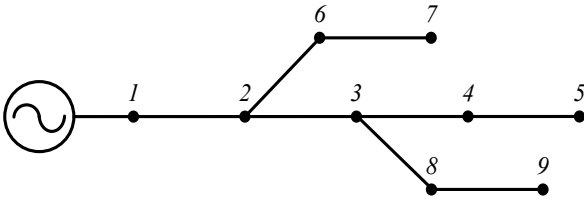


Fig. 3. Test distribution network

The test is divided into three scenes:

- a) Not connected to photovoltaic
- b) Node 2 connected to photovoltaic
- c) Node 3 connected to photovoltaic

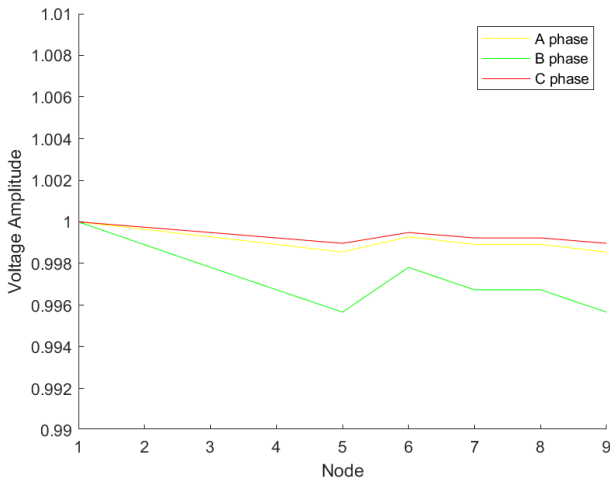
Calculate the unbalanced power flow and obtain the result in Figure 4(a). Due to the connection of a larger load, the voltage amplitude drop of phase B is significantly greater than that of phase AC.

Connect photovoltaic power to nodes 2 and 3 respectively, and the results are shown in Figure 4(b) and Figure 4(c).

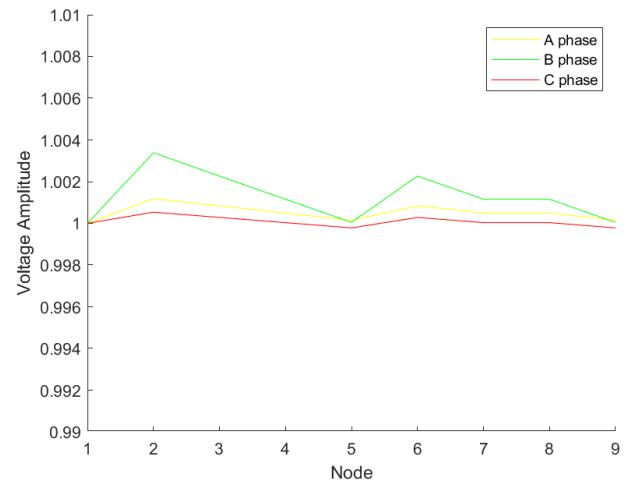
TABLE II. ITERATIONS OF DIFFERENT SCENES

Scenes	Connect to photovoltaic system or not	Number of iterations
Scene a	no	4
Scene b	yes	4
Scene c	yes	4

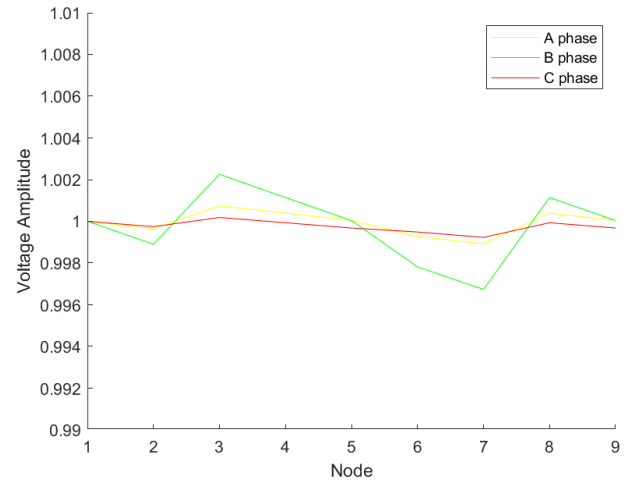
It can be seen that the forward and backward sweep method considering the integration of photovoltaic does not reduce its effectiveness and convergence speed.



(a) Voltages of scene a



(b) Voltages of scene b



(c) Voltages of scene c

Fig. 4. Calculation process of each method

V. CONCLUSION

In this paper, a three-phase unbalanced power flow algorithm based on traditional forward and backward sweep method is proposed which can solve power flow in distribution system with photovoltaic system. The algorithm has been simulated in test system and the simulation results show good convergence.

VI. ACKNOWLEDGMENT

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